

Assignment 5a/b - Design Patterns

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Requirements - Pet Shelter Management System

Below are functional requirements for the pet shelter management application. **You do not have to fulfill all requirements** - these are ideas to help guide your implementation. You must fulfill at least six requirements total. Your simulation should stay in this "pet shelter" realm but you can be creative with your ideas about what exactly you want to do.

Pet Shelter Simulation

Numbering and flexibility: Use R1, R2, etc. and sub-bullets like R1.2 when documenting. You do **not** have to implement every sub-bullet; implement a reasonable subset and explain your adaptations.

1. R1. Core Animals and Intake:

- a) Start with at least 5 animals and track basic attributes (ID, species, age, health status).
- b) New animals can arrive during the simulation (stray, surrender, transfer).
- c) Maintain a simple status for each animal (e.g., intake, available, pending, adopted).

2. R2. People and Assignments:

- a) Define at least two staff roles (e.g., veterinarian, technician, counselor).
- b) Each staff member has a capacity and can be assigned to animals or tasks.
- c) Staff availability can change over time (simple shift or on/off availability).

3. R3. Adoption and Matching:

- a) Adopters have basic preferences (species, size, age range, temperament).
- b) Match adopters to animals and update adoption status (available, pending, adopted).
- c) Handle at least one adoption outcome (successful or failed/returned).

4. **R4. Care and Medical Events:**

- a) Provide basic care actions (feeding, cleaning, exercise, or socialization).
- b) Schedule a simple medical action (e.g., intake exam or vaccination).
- c) Update animal health or notes when care/medical actions occur.

Resources and Simulation (MUST HAVE do not count as the 6 requirements):

1. Run the simulation in cycles (each cycle could represent a day).
2. Output clear state changes each cycle e.g.: new intakes, assignments, adoptions, or care events.
3. Simulate at least one week.

Note: Singleton pattern is NOT permitted as one of your two required patterns (it's too simple). You may use Singleton for utility purposes but it won't count toward your pattern requirement.

EXAMPLE

Example of Six Requirements You Could Fulfill: You do **not** need to implement every sub-bullet of a requirement to count it. You can implement a reasonable subset and adapt it, as long as you clearly document what you did. For example, a valid set of six could be:

- **R1.1** Start with at least 5 animals that have species, age, and ID.
- **R1.2** New animals can arrive once per day and follow the same basic attributes.
- **R2.1** Include an employee who handles adoptions and an employee who handles animal care.
- **R3.1** Adopters choose animals based on preferences and availability.
- **R3.3** Handle adoptions (successful or returned).
- **R4.1** Basic feeding and cleaning happens at least twice per cycle.

Example Pattern Pairings for the above chosen 6 example requirements

- **Factory + State:** Use a Factory to create animals or staff (R1, R2), and a State pattern to manage animal status transitions like `intake` → `available` → `pending` → `adopted` (R1, R3).

Remember: The key is to choose two patterns that work together and complement each other in solving your pet shelter system's challenges. Document clearly why you chose them!